

## AFJORD II, Norway

WMO: 012420

Lat: 63.966N

Lon: 10.216E

Elev: 30

StdP: 100.97

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 1             | -14.0      | -11.1 | -17.3                         | 0.8 | -11.4 | -15.2 | 1.0 | -9.1 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.9                    | 26.4            | 17.7 | 23.9 | 16.2 | 21.8 | 15.2 | 18.5                | 24.7 | 17.0 | 22.4 | 15.9 | 20.5 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.0                            | 11.4 | 21.0 | 14.7 | 10.5 | 19.4 | 13.6 | 9.8 | 17.7 | 52.5          | 24.6 | 47.9 | 22.6 | 44.5 | 20.7 | 23.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| N/A               | N/A  | N/A | -15.5                      | 29.3 | 2.4                | 2.2 | -17.2                                              | 30.9 | -18.7      | 32.3 | -20.0      | 33.5 | -21.8      | 35.2 |
|                   |      |     | -15.9                      | 19.9 | 2.3                | 1.7 | -17.6                                              | 21.2 | -19.0      | 22.2 | -20.3      | 23.2 | -22.0      | 24.4 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.3         | -1.5  | -0.7  | 1.2   | 4.9   | 9.2   | 12.0  | 15.0  | 14.4  | 11.4  | 6.3   | 1.9   | 0.3   |  |  |
|                                                                        | DBStd       | 6.91        | 4.67  | 4.45  | 3.81  | 3.39  | 3.90  | 3.25  | 3.69  | 2.83  | 2.88  | 3.27  | 4.30  | 4.37  |  |  |
|                                                                        | HDD10.0     | 1846        | 357   | 298   | 273   | 158   | 63    | 13    | 1     | 1     | 18    | 122   | 243   | 301   |  |  |
|                                                                        | HDD18.3     | 4433        | 615   | 532   | 531   | 402   | 284   | 191   | 119   | 126   | 208   | 374   | 492   | 559   |  |  |
|                                                                        | CDD10.0     | 478         | 0     | 0     | 0     | 6     | 38    | 74    | 157   | 138   | 59    | 6     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 23          | 0     | 0     | 0     | 0     | 1     | 2     | 16    | 5     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 266         | 0     | 0     | 0     | 0     | 14    | 27    | 177   | 47    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 58          | 0     | 0     | 0     | 0     | 1     | 3     | 52    | 3     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | N/A         | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1471        | 149   | 137   | 146   | 85    | 75    | 102   | 83    | 100   | 144   | 149   | 141   | 161   |  |  |
|                                                                        | PrecMax     | 1816        | 444   | 335   | 388   | 218   | 139   | 203   | 183   | 193   | 390   | 286   | 325   | 331   |  |  |
|                                                                        | PrecMin     | 1015        | 5     | 17    | 57    | 26    | 33    | 18    | 20    | 19    | 30    | 27    | 6     | 34    |  |  |
|                                                                        | PrecStd     | 212         | 96    | 73    | 78    | 49    | 26    | 41    | 41    | 48    | 92    | 52    | 93    | 73    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 8.0   | 7.9   | 11.2  | 20.1  | 25.1  | 26.5  | 30.7  | 27.1  | 22.3  | 18.0  | 11.5  | 9.1   |  |  |
|                                                                        |             | MCWB        | 5.2   | 5.0   | 5.6   | 11.3  | 14.7  | 16.0  | 19.8  | 17.1  | 14.9  | 12.6  | 9.0   | 6.6   |  |  |
|                                                                        | 2%          | DB          | 6.8   | 7.0   | 8.6   | 16.4  | 22.2  | 23.0  | 28.1  | 24.6  | 19.6  | 14.5  | 10.1  | 8.0   |  |  |
|                                                                        |             | MCWB        | 4.9   | 4.5   | 4.5   | 9.1   | 13.7  | 15.5  | 18.9  | 16.8  | 13.9  | 10.4  | 7.7   | 6.1   |  |  |
|                                                                        | 5%          | DB          | 5.5   | 6.1   | 7.2   | 13.6  | 19.4  | 20.5  | 24.9  | 22.2  | 17.9  | 12.4  | 8.7   | 6.9   |  |  |
|                                                                        |             | MCWB        | 3.7   | 4.0   | 4.6   | 7.7   | 12.5  | 13.8  | 17.7  | 15.5  | 13.2  | 8.9   | 6.6   | 4.9   |  |  |
|                                                                        | 10%         | DB          | 4.3   | 5.0   | 6.3   | 10.7  | 16.5  | 18.4  | 22.4  | 20.3  | 16.3  | 10.9  | 7.5   | 5.9   |  |  |
|                                                                        |             | MCWB        | 2.8   | 3.1   | 4.0   | 6.0   | 10.6  | 13.0  | 16.1  | 14.9  | 12.2  | 8.2   | 5.5   | 4.0   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.7   | 6.5   | 7.2   | 11.7  | 16.1  | 17.4  | 21.8  | 19.0  | 16.1  | 13.1  | 10.0  | 8.2   |  |  |
|                                                                        |             | MCDB        | 7.5   | 7.0   | 8.4   | 19.3  | 23.3  | 24.6  | 29.3  | 24.4  | 20.4  | 16.3  | 11.1  | 8.6   |  |  |
|                                                                        | 2%          | WB          | 5.5   | 5.5   | 6.3   | 9.5   | 14.2  | 15.8  | 19.7  | 17.4  | 14.8  | 10.9  | 8.4   | 6.7   |  |  |
|                                                                        |             | MCDB        | 6.2   | 6.3   | 7.3   | 15.6  | 21.0  | 21.7  | 26.3  | 22.8  | 18.1  | 13.9  | 9.6   | 7.3   |  |  |
|                                                                        | 5%          | WB          | 4.0   | 4.4   | 5.2   | 8.1   | 12.8  | 14.5  | 18.1  | 16.2  | 13.8  | 9.6   | 7.1   | 5.2   |  |  |
|                                                                        |             | MCDB        | 5.0   | 5.6   | 6.9   | 12.3  | 18.6  | 19.2  | 24.2  | 21.0  | 16.7  | 11.8  | 8.4   | 6.4   |  |  |
|                                                                        | 10%         | WB          | 2.8   | 3.1   | 4.1   | 6.8   | 11.3  | 13.3  | 16.6  | 15.2  | 12.8  | 8.6   | 5.6   | 3.9   |  |  |
|                                                                        |             | MCDB        | 4.1   | 4.8   | 6.2   | 9.7   | 16.0  | 17.6  | 21.5  | 19.4  | 15.6  | 10.4  | 7.1   | 5.6   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.7   | 5.3   | 6.9   | 8.6   | 9.3   | 8.5   | 8.9   | 8.8   | 7.4   | 6.3   | 4.9   | 5.0   |  |  |
|                                                                        |             | MCDBR       | 5.0   | 5.5   | 7.0   | 12.9  | 14.4  | 13.0  | 14.0  | 12.4  | 10.0  | 8.1   | 5.4   | 5.5   |  |  |
|                                                                        | 5% WB       | MCWBR       | 4.0   | 4.1   | 4.7   | 7.2   | 7.2   | 6.7   | 6.4   | 6.0   | 5.7   | 5.1   | 4.1   | 4.5   |  |  |
|                                                                        |             | MCWBR       | 4.7   | 5.1   | 5.9   | 11.2  | 13.1  | 12.1  | 13.5  | 10.8  | 9.1   | 6.9   | 5.3   | 5.6   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.391 | 0.306 | 0.333 | 0.358 | 0.355 | 0.356 | 0.363 | 0.351 | 0.337 | 0.313 | 0.358 | 0.362 |  |  |
|                                                                        |             | taud        | 2.640 | 2.207 | 2.279 | 2.351 | 2.429 | 2.459 | 2.465 | 2.494 | 2.510 | 2.396 | 2.557 | 2.493 |  |  |
|                                                                        | Ebn at Noon |             | 318   | 599   | 749   | 818   | 857   | 863   | 844   | 822   | 757   | 605   | 309   | 140   |  |  |
|                                                                        |             | Edh at Noon | 28    | 60    | 83    | 97    | 98    | 97    | 94    | 84    | 68    | 50    | 25    | 16    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.13  | 0.60  | 1.80  | 3.52  | 4.61  | 4.89  | 4.64  | 3.56  | 2.09  | 0.91  | 0.22  | 0.06  |  |  |
|                                                                        | RadStd      |             | 0.02  | 0.10  | 0.28  | 0.30  | 0.28  | 0.53  | 0.46  | 0.38  | 0.24  | 0.13  | 0.04  | 0.01  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (33 neighbors) | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | -143        | -189    | +67     | N/A     |

Nomenclature: See separate page